IMARCH LTD

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KNOWLEDGE ARCHITECTURE FOR A MEDICAL UNIVERSITY — Institutional Trust Architecture in Academic Publishing

Presentation Materials for UMSS

This document presents a next-generation knowledge architecture for a medical university, reconciling global scientific openness with absolute protection of clinical confidentiality. It sets out: a two-loop architecture (open science and secure vault), an automated typesetting pipeline, an RAG navigator, cryptographic copyright protection, and an institutional overlay for project financing and verification. The material is intended for restricted circulation within the strategic design of university publishing.

IMARCH Team

2026-08-27

Classification: RESTRICTED. Strictly Confidential

Knowledge Architecture for a Medical University

Presentation Materials for UMSS

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Key Theses:

- Two-loop architecture: Open Science (Loop A) and Secure Vault (Loop B).
- Automated typesetting pipeline — from raw text to finished monograph in seconds.
- RAG Navigator — an intelligent assistant for PDF-based research.
- Cryptographic Passport — copyright protection at SHA-256 level.
- Institutional overlay: verification, compliance, and UK legal umbrella.

Opening Thesis — The Challenges of Modern Medical Publishing

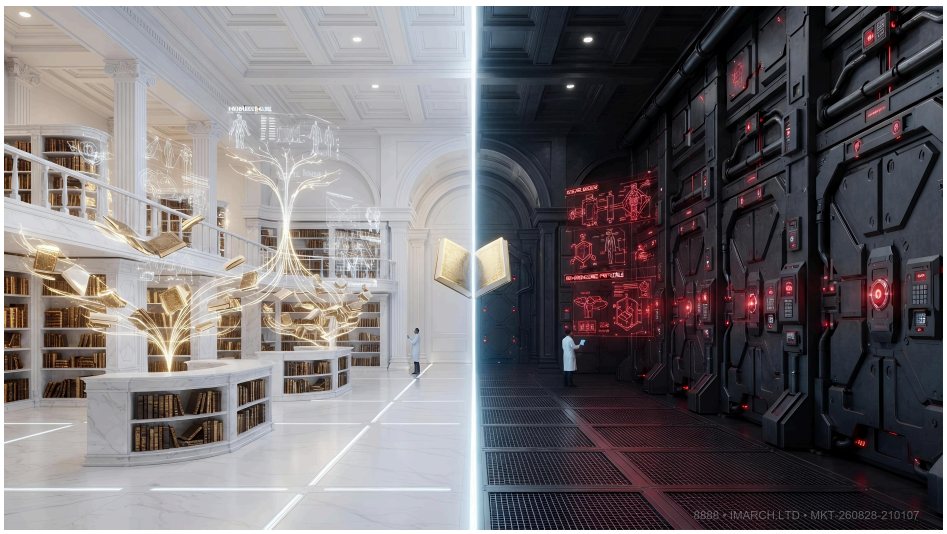
Dear Colleagues. IMARCH LTD presents a next-generation knowledge architecture for a medical university. We combine the irreconcilable: global scientific openness and absolute confidentiality of clinical data. Today, the academic world is burdened with routine: typesetting a single monograph can take up to six months, research remains invisible to the global scientific community, and confidential clinical data are at risk of leakage. We need a system that separates the streams: open science for the world, and a secure vault for clinical confidentiality.

The Chaos of Manual Labour in the Publishing Process

At present, the university publishing process is trapped in the paradigm of Word files and shared folders. Typesetting a single monograph can take up to six months. This is manual labour that does not scale and does not protect data. Tables lose alignment, formulas break formatting, references become tangled. Editorial staff spend 80% of their time on mechanical corrections rather than substantive editing. This is not merely inefficient — it is critically vulnerable to quality and data security failures.

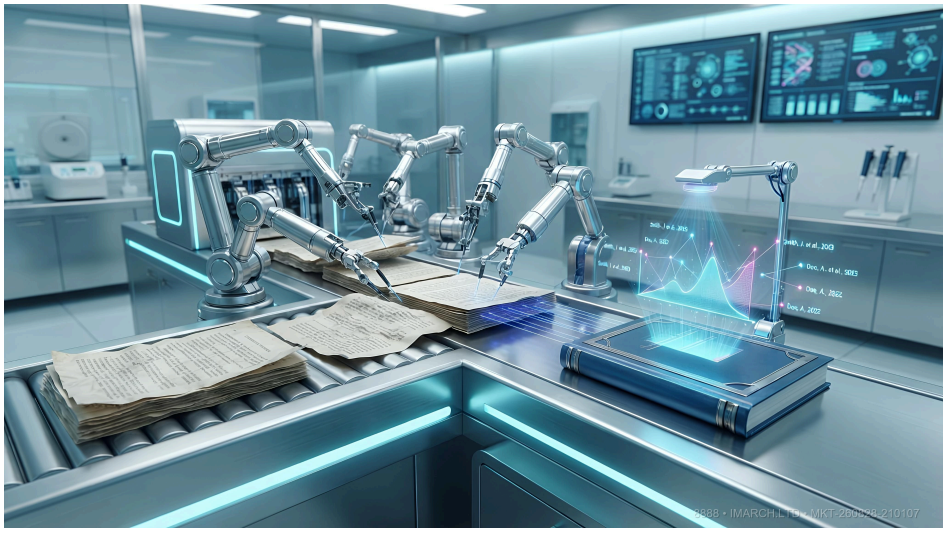
The Breaking Point — Leakage and Invisibility

This coin has two sides. On the outside, our work remains invisible to global science — it is not indexed by Google Scholar and is lost in the flood of publications. On the inside, confidential clinical data risk exposure, and patent integrity is called into question. This is a double blow: we lose reputation on the global stage while endangering medical confidentiality within. The gap between openness and security has become critical.



The Manifesto of Duality — Two-Loop Architecture

IMARCH offers a solution: separate the streams. We build a two-loop system. Loop 'A' operates for the external world — it drives scientometrics and institutional reputation. Loop 'B' operates internally — it is a vault that no one can penetrate. We create a digital perimeter where open publications become a driver of ranking, while protected data remain under robust security. This is not a compromise — it is an architectural decision.



Loop A — The Open Science Factory

The first loop is an open-science factory. We take raw professor's text and turn it into a typesetting masterpiece in seconds. The system automatically normalises references, aligns tables, and adapts formulas to a unified academic standard. Moreover, we embed search markup and an AI navigator, so that the monograph can be interrogated, not merely read. This transforms a passive PDF into an active intellectual resource.

Loop B — The Secure Vault

The second loop is a repository worthy of a GDPR-level data bank. Every file passes through role-based access control, SHA-256 cryptographic encryption, and full audit logging. Substitution or leakage becomes technically impossible. Access is strictly controlled via biometrics and digital keys. This is not merely a folder of documents — it is a digital vault with a complete audit trail of every action.

Pipeline Instead of Routine — Automated Typesetting

We eliminate the human factor from mechanical work. Our pipeline accepts any file — whether Word or LaTeX — and brings it to a single academic standard. The system automatically verifies references, aligns tables, formats bibliographies, and generates indices. Editorial staff cease to be typesetters and become scientific editors. Their expertise is redirected towards content, not mechanical corrections.

RAG Navigator — An Intelligent Assistant

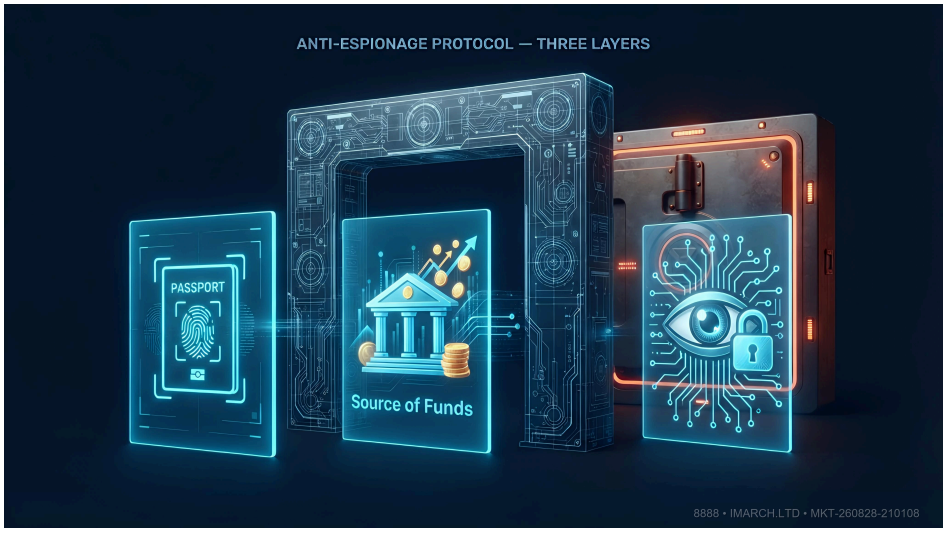
We turn a dead PDF into an intelligent assistant. A student or doctor opens the site and asks: 'What treatment protocols are described in Chapter 4?' The system instantly returns an answer with the exact page reference. This is not keyword search — it is semantic understanding. The neural network analyses the content, builds semantic connections, and provides a contextual answer strictly within the document, without hallucination. The monograph becomes a living dialogue.

Cryptographic Passport — Copyright Protection

Copyright must be ironclad. We assign a unique ID to every article. The document's hash is embedded in the metadata. If anyone changes so much as a comma, the system will instantly detect the alteration. This creates an absolute evidential basis for patent disputes and intellectual property protection. The author receives a digital certificate that cannot be forged.

Institutional Overlay — The Verification Model

Financing and IP protection are built on a cryptographic verification mechanism. The developer deposits specifications via the Telegram-Gate, receives a SHA-256 hash and a Trade Secrets Certificate from IMARCH LTD. The project undergoes double-blind independent peer review. This eliminates human bias and corruption risks. Every milestone is verified, and investments are protected against misuse.



IP Protection and Access Control — Three Layers

IP protection is built on a three-layer compliance framework. Layer A — KYC/AML & Sanctions Screening: UBO checks against OFAC, UK Sanctions List, EU Consolidated List, and PEP databases. Layer B — Source of Funds: verification of capital legitimacy through G7/FATF-compliant banks. Layer C — Zero-Knowledge Protection: the investor sees only milestone statuses and expert conclusions, with no access to engineering details. Documentation never leaves the sovereign perimeter.

Legal Umbrella — International Validation

The project is protected by a set of international legal instruments. The UK Trade Secrets Regulations 2018 safeguard commercial confidentiality. The Electronic Communications Act 2000 gives legal force to DKIM signatures. The EU eIDAS 2.0 Regulation recognises blockchain-sealed hashes as electronic timestamps with absolute evidential value. The system is fully valid under the jurisdictions of the United Kingdom, the European Union, and international arbitration.

Connection to the IMARCH Ecosystem

The project is integrated into the full institutional contour of the IMARCH ecosystem. Chronicle OS Journal publishes specifications and monographs. Verdictum provides cryptographic IP fixation and tranche-based escrow. IMARCH LTD (UK) supplies legal protection. TFM Engine handles documentation typesetting. Q-HUB-STUDIO generates visual materials for tenders. Everything works as a unified pipeline — from idea to serial deployment.

Paradigm Comparison — Old and New Approaches

Let us compare. The traditional approach means months of waiting and grey stacks of paper. The IMARCH approach means seconds of assembly, global visibility, and a vault that cannot be picked. We transform the publishing house from a cost centre into an intellectual factory, where every document is an asset, not an archive.

The Pilot Project

We propose to start small. Give us one manuscript, one monograph, and one set of clinical data. Within 24 hours, we will show you not a presentation, but a working prototype of your future publishing house. This is not a promise — it is an offer. We will prove our effectiveness on a real case. After that, you will see for yourself how the knowledge architecture transforms your university.

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KNOWLEDGE ARCHITECTURE AND VERIFICATION PROTOCOL FOR THE ACADEMIC COUNCIL (UMSS)

Circuit Registration Number: RN-0043+ Document Status: Official Technological Regulation and RAG Appendix for In-Depth Study (“for reading with a pencil”)

1. ABOUT THE PROJECT AND THE IMARCH TEAM

1.1. Who We Are

IMARCH LTD (UK) is a company registered in England and Wales (Company No. 16791765). We specialise in institutional trust architecture for governmental, academic, and medical organisations. Our profile is not that of a "printing house" or an "IT outsourcer", but rather an R&D partner and systems integrator that builds publishing ecosystems on the level of leading world publishers (Nature, Elsevier), with a focus on speed, transparency, and cryptographic protection.

1.2. Project Purpose for UMSS

To transfer the entire scientific and educational-methodological document flow of the university from manual (Word + shared folders) to an industrial pipeline with automated typesetting, semantic indexing, protected storage of clinical data, and transparent audit of every action.

1.3. Key Principles

- Two-loop design – separation of open science and closed medical confidentiality.
- Non-invasiveness – preservation of authors' habitual working formats (Word, LaTeX) without compulsory retraining.
- Legal validity – all processes are underpinned by British law (UK Trade Secrets, eIDAS, SHA-256).
- Scalability – from a pilot monograph to a full-scale publishing house with dozens of journals.

2. SYSTEMIC PAINS OF THE CURRENT STATE

Analysis of the university's current state reveals four systemic problems:

1. Manual typesetting and lengthy timelines Editorial and Publishing Department staff spend up to 80% of their time on mechanical formatting of styles, tables, formulas, and bibliographies. A monograph of 300 pages requires 3 to 6 months with 2–3 people working full-time. With staff turnover, standards become blurred and quality declines.
2. Invisibility to the global scientific community Finished PDF files are placed on local websites without semantic markup (Dublin Core, Schema.org, OpenGraph). As a result, Google Scholar, Scopus, and ResearchGate do not properly index the materials – citations drop, the Hirsch index stagnates, and the university's international recognition does not grow.
3. Risks of leakage and regulatory non-compliance Storage of clinical data, test results, and patients' personal data in shared network folders, flash drives, or unsecured clouds creates a direct threat. The absence of access control (RBAC) and action auditing makes compliance with GDPR and the UK Data Protection Act 2018 impossible, exposing the university to fines of up to €20 million.
4. Legal vulnerability of authorship and patent purity The absence of immutable cryptographic timestamps and digital passports for documents deprives the university of the ability to prove priority in patent disputes, and also creates risks in peer review and international data exchange.

3. TWO-LOOP ARCHITECTURE (DUAL-CIRCUIT)

The uniqueness of the IMARCH proposal lies in the separation of publishing activities into two sealed compartments within a single technological core.

3.1. Loop A (Public Node) – "Open Science"

- Purpose: maximisation of reach, citation, and scientometric indicators of the university.
- Input: any file (docx, tex, md) is uploaded via a web interface or Telegram bot.
- Process:
  - Normalisation of styles (headings, fonts, indentation) using the modern typesetting engine Typst.
  - Automatic bibliography generation (BibTeX, EndNote).
  - Embedding of metadata (Dublin Core, Schema.org, OpenGraph) for SEO and proper indexing.

  - Generation of an interactive RAG widget (chat-bot) based on the full text, allowing readers to ask questions and receive answers with page references.
- Result: a finished PDF and web version in less than 24 hours, indexing in Google Scholar within 48 hours, increased visibility and citation.

3.2. Loop B (Secure Vault) – "Closed Perimeter"

- Purpose: protection of medical confidentiality, clinical cases, unpublished patents, and financial documents.
- Architecture: isolated cloud storage on AWS/GCP with SOC2 and GDPR certification, physically located in the United Kingdom (or on-premise at the customer's request).
- Access management:
  - Role-based model (RBAC) – Rector, Vice-Rectors, Heads of Departments, laboratory staff, external auditors.
  - Two-factor authentication (TOTP or biometrics).
  - Full logging of every action (viewing, editing, downloading, transfer).
- Cryptographic protection:
  - Every uploaded document receives a SHA-256 hash and a digital passport with a unique identifier.
  - Document versions are recorded as new hashes with immutable timestamps.
  - Any attempt at substitution or deletion is immediately detected by the system and made available for audit.

4. TECHNOLOGY STACK AND SPEED

4.1. Markup Language – Markdown

- Authors write in plain text (or submit Word – we convert automatically).
- Easy to embed formulas (LaTeX syntax), tables, images, cross-references.

4.2. Typesetting Engine – Typst

- Modern functional language for typesetting, surpassing LaTeX in speed and usability.
- Compilation of a 100-page document with formulas and references takes <5 seconds.
- Support for custom styles (UMSS brand book) – all publications will be maintained in a unified corporate design.

4.3. Processing Pipeline

- File intake via web interface or Telegram bot.
- Normalisation: docx → markdown conversion, style cleansing.
- Validation of bibliographic and cross-references.
- Auto-generation of table of contents, list of illustrations, index.
- Finalisation: export to PDF, ePub, HTML and upload to repository.

4.4. Guaranteed Timelines

- Basic assembly (up to 300 pages) – within 24 working hours from receipt of valid source materials.
- Full cycle with peer review – up to 5 working days (depending on reviewer speed).
- For archival materials (conversion of old issues) – timelines are discussed separately.

4.5. Training and Adaptation

- Editorial and Publishing Department staff undergo training in using the control panel, quality control, and metadata configuration (2–3 days).
- For authors, brief instructions are provided for formatting in Word (using ready-made templates) or Markdown.
- Administrators receive access to a dashboard for rights management and monitoring.

5. COLLABORATION POLICY AND COMMERCIAL BLOCK

5.1. Wholesale-Partnership Basis

We operate on a scheme of comprehensive wholesale supply of publication flows and infrastructure solutions. This means we do not sell individual licences or one-off services, but rather build long-term partnerships for processing the entire body of scientific and educational materials of the university.

5.2. Technological Support

From each realised flow (journal, monograph, textbook), 30% of the funds are directed to the software developer for technical support, maintenance, and development of the circuit. This ensures continuous system updating, adaptation to new standards, and rapid resolution of any abnormal situations.

5.3. SaaS Infrastructure

Cloud SaaS solutions are considered a strategic prospect for the future, as the system is deployed and field-tested. At the initial stage, we offer a hybrid model (on-premise + private cloud) to guarantee full control over data and compliance with regulatory requirements.

5.4. Procedure for Financial Discussion

Financial terms, detailed calculations, and economic parameters are discussed exclusively with the university's leadership after the signing of a Non-Disclosure Agreement (NDA) and legal formalisation of the cooperation framework. For its part, the company employs advanced marketing technologies and strives to care for partners so that cooperation is maximally beneficial and transparent.

6. IMPLEMENTATION ROADMAP (PILOT PROJECT)

Stage 1. Audit and Deposit of Intellectual Property (1 week) At this stage, we collect and analyse the university's current materials, assess the existing infrastructure and documentation. In parallel, a Non-Disclosure Agreement (NDA) and a Trade Secrets Agreement are signed, allowing for legally protected data exchange. A unique V-ID identifier is generated in the Verdictum registry, becoming the cryptographic passport for the entire project. Result (Milestone): Cryptographic project passport obtained.

Stage 2. Pilot Monograph Assembly (2 weeks) We receive one manuscript from a specialised department (e.g., clinical or theoretical medicine). The system automatically performs typesetting using the Typst engine, then creates an RAG widget based on the text — an intelligent assistant for searching answers within the document. The finished interactive layout is presented to the Academic Council to demonstrate capabilities and address questions. Result (Milestone): Completed interactive mock-up (prototype).

Stage 3. Deployment of the Scientific Journal Infrastructure (1 month) At this stage, we configure the server environment (cloud or on-premise), install and configure the Typst pipeline for automatic assembly, deploy the role-based access control system (RBAC), and integrate it with the existing UMSS authentication system (LDAP, Shibboleth, or equivalent). Concurrently, training is provided for Editorial and Publishing Department staff on working with the control panel, quality control, and metadata. Result (Milestone): Working stand for receiving manuscripts.

Stage 4. Pilot Journal Issue (1 month) At this stage, we accept 10–15 manuscripts from authors, conduct the peer-review process (including using a double-blind peer-review system), automatically perform typesetting, and publish a full journal issue. After publication, we measure citations, analyse indexing in search engines, and collect feedback from the editorial board and authors. Result (Milestone): First pilot issue published.

Stage 5. Scaling to All Publications and Industrial Operation (3 months) At the final stage, we migrate all existing journals, monograph series, and educational-methodological complexes of the university to the new platform. Integration with international repositories (Scopus, Web of Science, CrossRef) is performed for

automatic DOI registration and updating of scientometric indicators. The system is handed over for industrial operation with a full package of documentation, regulations, and trained personnel. Result (Milestone): Full transition of all publications and entry into permanent operation.

7. ECONOMIC EFFICIENCY AND COLLABORATION TERMS

Financial terms, detailed calculations, and economic parameters are discussed exclusively with the university's leadership after the signing of a Non-Disclosure Agreement (NDA) and legal formalisation of the framework for further cooperation.

For its part, the company employs advanced marketing technologies and strives to care for partners so that cooperation is as beneficial as possible.

8. LEGAL BLOCK AND DEPOSITION

This document is an official demonstration artefact and a presentation projection of the institutional standard of the IMARCH / Chronicle OS ecosystem.

- Document Identifier: `IMARCH-STD-IPP-001` (Registration Circuit: `RN-0043`).
- Legal Regime: Open Standard (CC BY-SA 4.0). Intellectual Domain of IMARCH LTD (UK).
- Verification and Immutability: The integrity and immutability of the substantive part are fixed by a cryptographic hash key (SHA-256) in the Verdictum system registry.
- Access Procedure: This document constitutes an autonomous node of a comprehensive package. The full array of primary specifications, source typesetting codes, and RAG indices is available upon request through the official comprehensive artefact passport based on the session node marker.

9. ADDITIONAL MATERIALS AND FAQ

9.1. Answers to Frequently Asked Questions

Question: "Won't our old archives be lost?" Answer: No. We provide tools for converting old documents (doc, pdf) into the new format. For archival issues, a static digital library can be created without RAG functionality, but with search and DOI.

Question: "We won't be able to retrain all the teachers" Answer: This is not required. Authors may continue to write in Word – our system converts the files. We also provide a simple template and video instructions. For those who wish to write in Markdown, there is a short guide.

Question: "What if the system breaks?" Answer: All components are containerised (Docker) and deployed in a cluster with automatic recovery. Weekly backup to a separate storage. Support 24/7 for the first 3 months.

Question: "Is this certified for medical data?" Answer: Yes, the architecture is designed with consideration of GDPR and UK Data Protection Act requirements. UK jurisdiction adds an additional layer of protection. If necessary, we can undergo an independent audit.

Question: "Can we start with just one journal?" Answer: Yes, this is the recommended path – a pilot project. After a successful launch, you will be able to scale the system to any other publications.

10. CONCLUSION

IMARCH offers UMSS not merely software, but an institutional trust architecture that moves university publishing into the digital age without losing control over data and without increasing staff. We combine the speed of industrial typesetting, the transparency of cryptographic audit, and the legal protection of British law. The pilot project can be launched within a month, and the effect will be felt in the very first issue.

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